Chapter 17, Problem 30P

Problem

(a) if *f*(*t*) is an even function, show that

$$c_n = \frac{2}{T}\int_0^{T/2} f(t)\cos n\omega_o t\,dt$$

(b) If *f*(*t*) is an odd function, show that

$$c_n = -\frac{j2}{T}\int_0^{T/2} f(t)\sin n\omega_o t\,dt$$

Step-by-step solution

Step 1 of 2

$$C_n = \frac{1}{T}\int_o^T f(t)e^{-jn\omega_o t}dt$$

$$C_n = \frac{a_n - jb_n}{2}$$

since we hav, $\cos n\omega_o t = \frac{1}{2}\left[e^{jn\omega_o t} + e^{-jn\omega_o t}\right]$

and $\sin n\omega_o t = \frac{1}{2j}\left[e^{jn\omega_o t} + e^{-jn\omega_o t}\right]$

Step 2 of 2

we have

$$f(t) = a_o + \sum_{n=1}^{\infty}\left[a_n \cos n\omega_o t + b_n \sin n\omega_o t\right]$$

$$a_o + \sum_{n=1}^{\infty} \frac{a_n e^{jn\omega_o t} + a_n e^{-jn\omega_o t}}{2} + \frac{b_n e^{jn\omega_o t} - b_n e^{-jn\omega_o t}}{2j}$$

$$a_o + \sum_{n=1}^{\infty} (a_n - jb_n)\frac{e^{jn\omega_o t}}{2} + (a_n + jb_n)\frac{e^{-jn\omega_o t}}{2}$$

$$let \quad C_n = \frac{a_n - jb_n}{2} \, and \quad C_{(-n)} = C_n^* = \frac{a_n + jb_n}{2}$$

$$f(t) = C_o + \sum_{n=1}^{\infty} C_n e^{jn\omega_o t} + C_n e^{-jn\omega_o t}$$

$$f(t) = \sum_{n=1}^{\infty} C_n e^{jn\omega_o t}$$

For even function

$$f(t) = a_o + \sum_{n=1}^{\infty} a_n \cos n\omega_o t$$

$$C_n = \frac{a_n - jb_n}{2} = \frac{a_n}{2}$$

$$C_n = \frac{4}{2 \times T}\int_o^{\frac{T}{2}} f(t) \cos n\omega_o t \, dt$$

$$= \frac{2}{T}\int_o^{\frac{T}{2}} f(t) \cos n\omega_o t \, dt$$

For odd function

$$f(t) = \sum_{n=1}^{\infty} b_n \sin n\omega_o t$$

$$b_n = \frac{4}{T}\int_o^{\frac{T}{2}} f(t) \sin n\omega_o t \, dt$$

$$C_n = \frac{a_n - jb_n}{2} = \frac{-jb_n}{2}$$

$$\boxed{= \frac{-2j}{T}\int_o^{\frac{T}{2}} f(t) \sin n\omega_o t \, dt}$$